



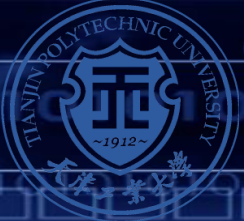
实验 源NAT— NAPT 配置



目 标

❖ 学完本章内容后，同学们将能够：

- 掌握 NAPT 的工作原理
- 掌握 NAPT的应用场景
- 掌握 NAPT的配置方法



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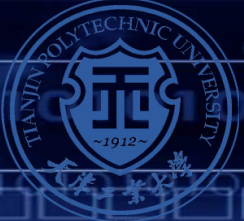
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配置NAPT



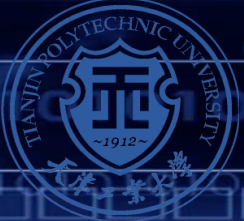
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实验目的



实验目的

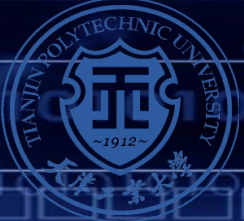
- ❖ 掌握 NAPT 的工作原理
- ❖ 了解NAPT 的应用场景
- ❖ 掌握NAPT 的配置方法



NAPT 工作原理

- ❖ 一种基于源IP地址NAT转换技术
- ❖ 同时实现接口的转换
- ❖ 将不同的内部地址映射到同一公有地址的不同端口号上，实现多对一地址转换。主要利用NAPT技术实现多对一地址转换。

192.168.1.1	→	155.133.87.1: 7111
192.168.1.2	→	155.133.87.1: 7112
192.168.1.3	→	155.133.87.1: 7113



NAPT 应用场景

- ◆ NAPT主要应用于实现私网中同时访问Internet的用户大于公网IP地址的场景。
- ◆ 比较常用的一种基于源地址转换的NAT技术
- ◆ 可以实现内网转外网，也可以反方向转换



2

实验网络结构

实验网络结构图

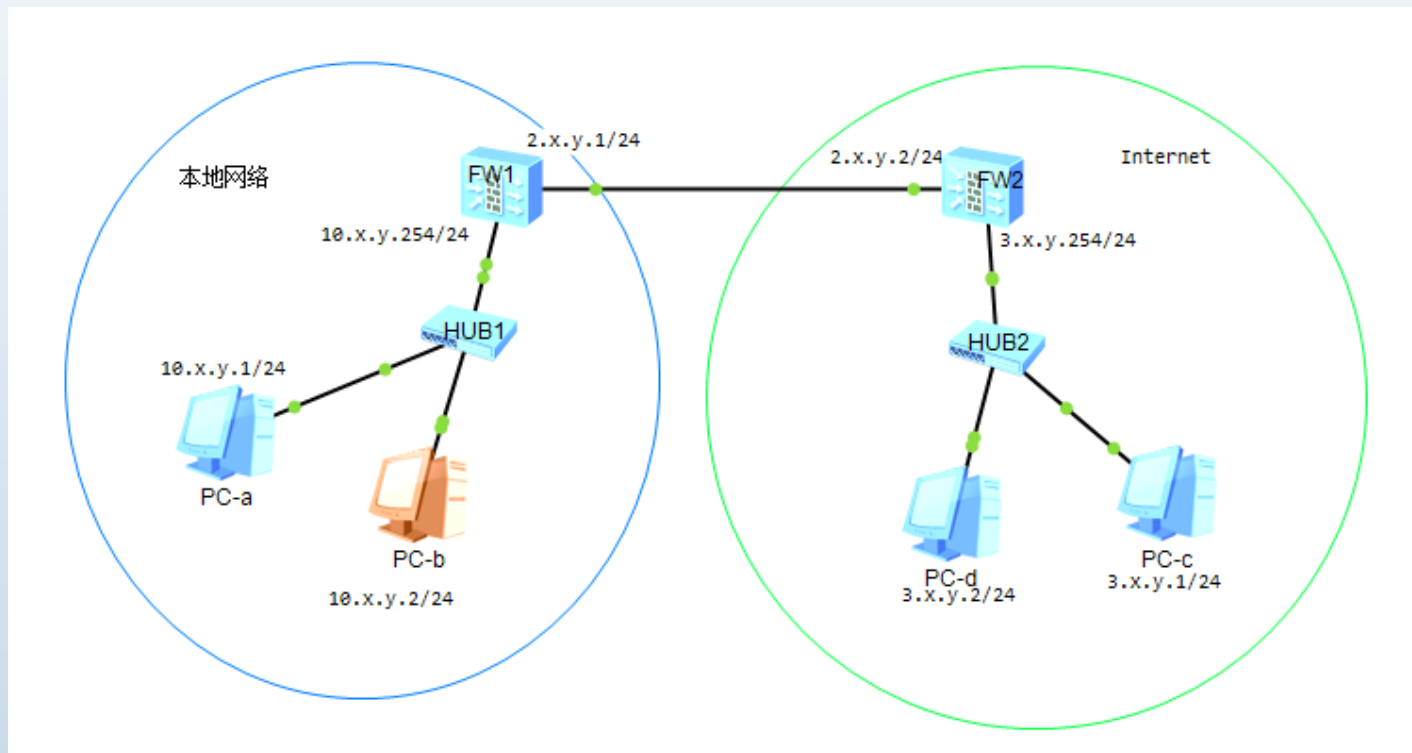
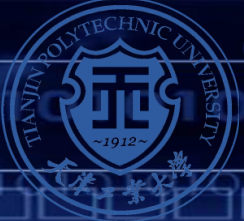
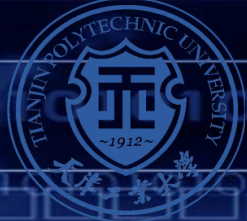


图1 实验网络结构图



组网需求

某公司在网络边界处部署了FW作为安全网关。为了使私网中10. x.y.0/24网段的用户可以正常访问Internet，需要在FW上配置源NAT策略。除了公网接口的IP地址外，公司还向ISP申请了3个IP地址（2.x.y.11 ~ 1. x.y.13）作为私网地址转换后的公网地址。网络如图1所示，其中FW2是ISP提供的接入网关。同时，要求私网用户与Internet上的主机之间能互访。

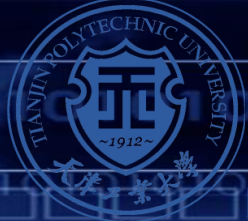


数据规划

网络设备名称	接口	IP地址	所属安全区域	互联对端设备	对端接口
FW1	G1/0/0	2.x.y.1/24	untrust	FW2	G1/0/0
	G1/0/1	10.x.y.254/24	trust	HUB1	e0/0/0
FW2	G1/0/0	2.x.y.2/30	untrust	FW2	G1/0/0
	G1/0/1	3.x.y.254/24	trust	HUB2	e0/0/0
HUB1	e0/0/0	无	无	FW1	G1/0/1
	e0/0/1	无	无	PC-a	
	e0/0/2	无	无	PC-b	
HUB2	e0/0/0	无	无	FW2	
	e0/0/1	无	无	PC-c	
	e0/0/2	无	无	PC-d	
主机名称	IP地址	子网掩码	默认网关		
PC-a	10.x.y.1	255.255.255.0	10.x.y.254		
PC-b	10.x.y.2	255.255.255.0	10.x.y.254		
PC-c	3.x.y.1	255.255.255.0	3.x.y.254		
PC-d	3.x.y.2	255.255.255.0	3.x.y.254		

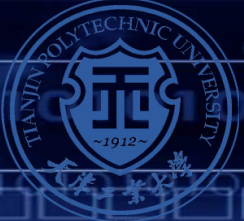
特别注意：IP地址中的x,y的含义

x：学号倒数第三位； y:学号倒数后两位



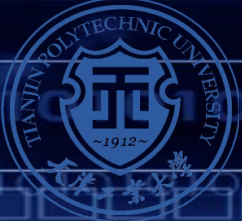
接口规划

设备名称	项目	数据	说明
防火墙 FW	g 1/0/1	IP地址：10.x.y.254/24 安全区域：trust	私网主机需要将192.168.1.1配置为默认网关
	g 1/0/0	IP地址：2.x.y.1/30 安全区域：Untrust	实际配置时需要按照ISP的要求 进行配置。
	允许访问Internet的私网网段	10.x.y.0/24	2.x.y.11-2.x.y.13
	转换后的公网地址	2.x.y.11-2.x.y.13	由于私网地址比公网地址多，无法做到地址一一映射，所以需要 开启允许端口转换，通过端口转 换实现公网地址复用。
	缺省路由	目的地址：0.0.0.0 下一跳：2.x.y.2	为了内网服务器对外提供的服 务流量可以正常转发至ISP的路 由器，可以在FW上配置去往 Internet的缺省路由。
	黑洞路由	目的地址：2.x.y.11 下一跳：NULL 0 目的地址：2.x.y.12 下一跳：NULL 0 目的地址：2.x.y.13 下一跳：NULL 0	为了避免外网用户访问Global地 址但没有匹配到Server-Map的报 文，在FW和Router之间形成路 由环路。



实验步骤

- ❖ 在私网服务器上配置缺省网关，使私网主机访问Internet时，将流量发往FW。
- ❖ 在R1上配置接口地址、静态路由，使从Internet返回的流量可以被正常转发至FW配置设备基本信息
- ❖ 配置防火墙参数
 - 配置接口IP地址和安全区域，完成网络基本参数配置。
 - 配置安全策略，允许私网指定网段与Internet进行报文交互。
 - 配置NAT地址池，配置时开启允许端口转换，以实现公网地址复用。
 - 配置源NAT策略，实现私网指定网段访问Internet时自动进行源地址转换。
 - 在FW上配置缺省路由，使私网流量可以正常转发至ISP的路由器。
 - 在FW上配置黑洞路由，防止环回路由



在私网主机上配置缺省网关

以PC1为例进行配置，参数参照接口数据规划表

PC-a

基础配置 命令行 组播 UDP发包工具 串口

主机名: pc-a

MAC 地址: 54-89-98-79-7F-77

IPv4 配置

☒ 静态 ☐ DHCP ☐ 自动获取 DNS 服务器地址

IP 地址: 10 . 1 . 1 . 1

子网掩码: 255 . 255 . 255 . 0

网关: 10 . 1 . 1 . 254

DNS1: 0 . 0 . 0 . 0

DNS2: 0 . 0 . 0 . 0

IPv6 配置

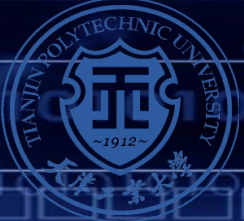
☒ 静态 ☐ DHCPv6

IPv6 地址: ::

前缀长度: 128

IPv6 网关: ::

应用



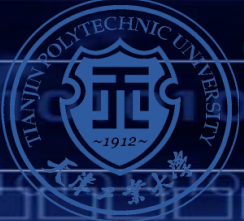
配置防火墙FW1

❖ 修改设备名称

- sysname 学号后三位-1

❖ 配置接口地址

- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.1 255.255.255.252(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 10.x.y.254 255.255.255.0
- service-manage ping permit



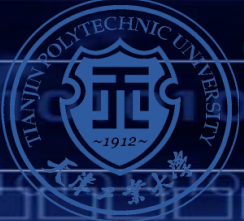
配置防火墙FW1

❖ 配置安全区域

- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0

❖ 配置缺省缺省路由

- ip route-static 0.0.0.0 0.0.0.0 2.x.y.2



防火墙FW1配置

❖ 配置安全策略：trust-untrust (实现trust区域与untrust区域通信)

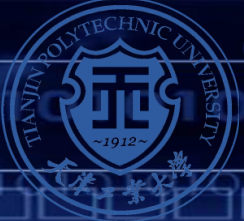
```
security-policy  
rule name trust-untrust  
source-zone trust  
destination-zone untrust  
source-address 10.x.y.0 mask 255.255.255.0  
action permit
```

❖ 配置缺省安全策略

- security-policy
- default action deny

❖ 保存配置

- 在普通用户模式下，save



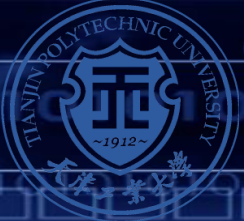
配置防火墙FW2

❖ 修改设备名称

- sysname 学号后三位-2

❖ 配置接口地址

- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.2 255.255.255.252(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 3.x.y.254 255.255.255.0
- service-manage ping permit



配置防火墙FW2

❖ 配置安全区域

- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0

❖ 配置缺省安全策略

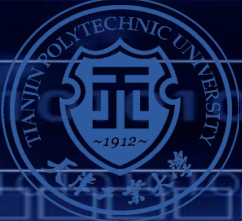
- security-policy
- default action permit

❖ 配置缺省缺省路由

- ip route-static 0.0.0.0 0.0.0.0 2.x.y.1

❖ 保存配置

- 在普通用户模式下，save

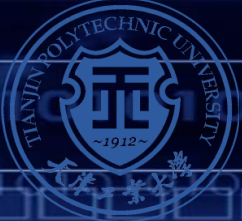


阶段性结果检查

❖ 查看FW1上接口配置信息

```
[test-1]display ip interface brief
2024-03-26 13:47:33.640
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 6
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 6
```

Interface	IP Address/Mask	Physical
GigabitEthernet0/0/0	192.168.0.1/24	down
GigabitEthernet1/0/0	2.2.2.1/24	up
GigabitEthernet1/0/1	10.1.1.254/24	up
GigabitEthernet1/0/2	unassigned	down
GigabitEthernet1/0/3	unassigned	down
GigabitEthernet1/0/4	unassigned	down
GigabitEthernet1/0/5	unassigned	down
GigabitEthernet1/0/6	unassigned	down
NULL0	unassigned	up
Virtual-if0	unassigned	up



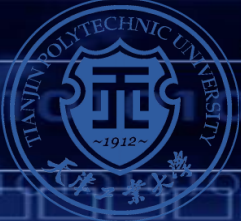
阶段性结果检查

- ❖ 网络中trust区域中符合安全策略要求的主机可以与untrust区域的主机通信
- ❖ PCa→网关 (ping)
- ❖ PCa→PCc(ping)

The image shows two side-by-side screenshots of the PC-a command prompt window in a network simulator. Both windows have tabs for '基础配置' (Basic Configuration), '命令行' (Command Line), '组播' (Multicast), 'UDP发包工具' (UDP Packet Tool), and '串口' (Serial Port). The '命令行' tab is active in both.

Left Screenshot: The user has entered the command `PC>ping 10.1.1.254`. The output shows four successful ping attempts to 10.1.1.254. Each attempt shows 32 data bytes, a sequence number (seq=1 to seq=4), a TTL of 255, and a response time of 63 ms, 15 ms, <1 ms, and <1 ms respectively. The statistics at the bottom show 4 packets transmitted and received, 0.00% packet loss, and a round-trip time of 0/19/63 ms.

Right Screenshot: The user has entered the command `PC>ping 3.3.3.1`. The output shows five successful ping attempts to 3.3.3.1. Each attempt shows 32 data bytes, a sequence number (seq=1 to seq=5), a TTL of 126, and a response time of 16 ms, 15 ms, <1 ms, 16 ms, and 16 ms respectively. The statistics at the bottom show 5 packets transmitted and received, 0.00% packet loss, and a round-trip time of 0/12/16 ms.



阶段性结果检查

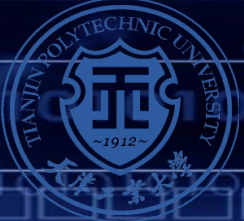
❖ 防火墙上观察会话表信息

- 有会话记录，且使用私网地址

```
FW1
[FW1]display firewall session table
2024-03-26 14:06:57.050
Current Total Sessions : 19
icmp VPN: public --> public 10.1.1.1:43220 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:44500 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:43988 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:45268 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:42964 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:43732 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:47572 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:46548 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:45780 --> 3.3.3.1:2048
icmp VPN: public --> public 10.1.1.1:44244 --> 3.3.3.1:2048
```

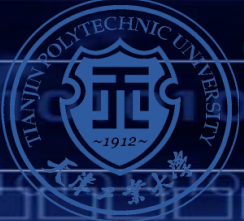
- 具体会话记录内容

```
[test-1]display firewall session table verbose
2024-03-26 14:09:27.570
Current Total Sessions : 20
icmp VPN: public --> public ID: c287f8bea8a1050726602d708
Zone: trust --> untrust TTL: 00:00:20 Left: 00:00:18
Recv Interface: GigabitEthernet1/0/1
Interface: GigabitEthernet1/0/0 NextHop: 2.2.2.2 MAC: 00e0-fcf7-7240
<--packets: 1 bytes: 60 --> packets: 1 bytes: 60
10.1.1.1:15829 --> 3.3.3.1:2048 PolicyName: trust-untrust
```



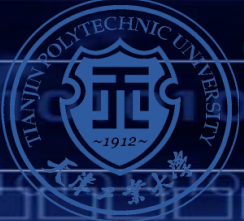
防火墙FW1上nat配置步骤

- ❖ Nat地址池
- ❖ Nat策略
- ❖ 配置黑洞路由



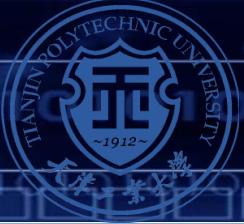
配置NAT 地址池

- ❖ **nat address-group addressgroup1**
- ❖ **mode pat**
- ❖ **section 0 2.x.y.11 2.x.y.13**
- ❖ **Route enable**



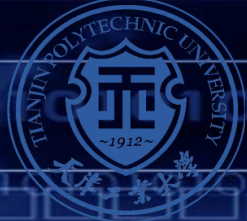
配置NAPT 策略

- ❖ nat-policy
- ❖ rule name policy_nat1
- ❖ source-zone trust
- ❖ destination-zone untrust
- ❖ source-address 10.x.y.0 24
- ❖ action source-nat address-group addressgroup1



配置黑洞路由

- ip route-static 2.x.y.11 32 null 0
- ip route-static 2.x.y.12 32 null 0
- ip route-static 2.x.y.13 32 null 0



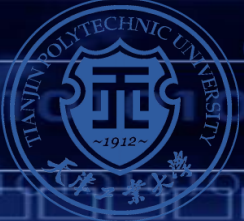
结果验证

❖ 查看路由表

display ip routing-table

```
[test-1]display ip routing-table
2024-04-01 03:49:42.720
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 9              Routes : 9

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
1/0/0      0.0.0.0/0    Static  60    0              RD   2.2.2.2         GigabitEthernet
2.2.2.0/24    Direct    0     0              D   2.2.2.1         GigabitEthernet
1/0/0      2.2.2.1/32   Direct    0     0              D   127.0.0.1       GigabitEthernet
1/0/0      2.2.2.11/32  Static   60    0              D   0.0.0.0         NULL0
2.2.2.12/32  Static   60    0              D   0.0.0.0         NULL0
1/0/1      10.1.1.0/24  Direct    0     0              D   10.1.1.254     GigabitEthernet
10.1.1.254/32 Direct    0     0              D   127.0.0.1       GigabitEthernet
1/0/1      127.0.0.0/8  Direct    0     0              D   127.0.0.1       InLoopBack0
127.0.0.1/32 Direct    0     0              D   127.0.0.1       InLoopBack0
```

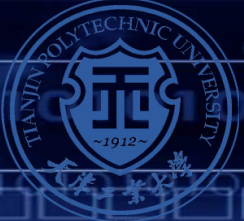


结果验证

❖ 查看nat 地址池

display nat address-group name addressgroup1

```
[test-1]display nat address-group name addressgroup1
2024-04-01 03:46:15.050
nat address-group addressgroup1 0
  reference count: 1
  mode pat
  status active
  route enable
  section 0 2.2.2.11 2.2.2.12
```

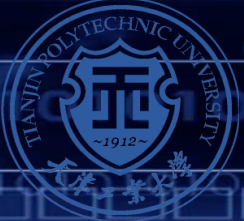


结果验证

❖ 查看nat 策略

display nat-policy rule name napt1

```
<test-1>disp nat-policy rule name napt1
2024-04-01 03:42:09.910
(10 times matched)
rule name napt1
source-zone trust
destination-zone untrust
source-address 10.1.1.0 mask 255.255.255.0
action source-nat address-group addressgroup1
```



结果验证

❖ PCa→PCc

结果显示：

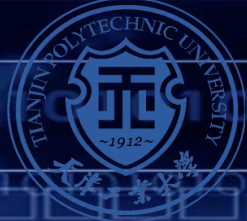
```
PC-a
基础配置  命令行  组播  UDP发包工具  串口

PC>ping 3.3.3.1

Ping 3.3.3.1: 32 data bytes, Press Ctrl_C to break
From 3.3.3.1: bytes=32 seq=1 ttl=126 time<1 ms
From 3.3.3.1: bytes=32 seq=2 ttl=126 time=16 ms
From 3.3.3.1: bytes=32 seq=3 ttl=126 time=15 ms
From 3.3.3.1: bytes=32 seq=4 ttl=126 time<1 ms
From 3.3.3.1: bytes=32 seq=5 ttl=126 time<1 ms

--- 3.3.3.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 0/6/16 ms

PC>
```



结果验证

❖ FW上查看NAT转换结果

```
[test-1] disp firewall session table
```

```
2024-04-01 03:28:29.310
```

```
Current Total Sessions : 3
```

```
icmp VPN: public --> public 10.1.1.1:48425[2.2.2.12:2052] --> 3.3.3.1:2048  
icmp VPN: public --> public 10.1.1.1:48169[2.2.2.12:2051] --> 3.3.3.1:2048  
icmp VPN: public --> public 10.1.1.1:47657[2.2.2.12:2050] --> 3.3.3.1:2048
```

```
[test-1] disp firewall session table verbose
```

```
2024-04-01 03:29:00.290
```

```
Current Total Sessions : 5
```

```
icmp VPN: public --> public ID: c487fca7df88790183d660a29f9
```

```
Zone: trust --> untrust TTL: 00:00:20 Left: 00:00:18
```

```
Recv Interface: GigabitEthernet1/0/1
```

```
Interface: GigabitEthernet1/0/0 NextHop: 2.2.2.2 MAC: 00e0-fcf7-7240
```

```
<--packets: 1 bytes: 60 --> packets: 1 bytes: 60
```

```
10.1.1.1:59433[2.2.2.12:2056] --> 3.3.3.1:2048 PolicyName: trust-untrust
```




谢谢！